BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: PR XI (02) 23

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** PR XI (02) 23

Note: Instructions (printed in Urdu on the sheet): 1. Each question has four circles; fill in only the circle of the correct answer. 2. Use a blue or black pen to fill the circles. 3. Filling more than one circle will be treated as a wrong answer.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The mass (in grams) of 3 moles of water is.....

(A) 90g (B) 196g

(C) 36g (D) 54g

Q2. The.....spectrum consists of dark lines in a bright background.

(A) Continuous (B) Line emission

(C) Line absorption (D) Visible

Q3. The appropriate condition applied in a discharge tube for the production of cathode rays is.....

(A) High voltage and low pressure (B) High voltage and high pressure

(C) Low voltage and low pressure (D) Low voltage and high pressure

Q4. Both VSEPR theory and VBT fail to show the unpaired electrons in.....molecule.

(A) H_2 (B) He

(C) O_2 (D) N_2

Q5. The triple bond in ethyne molecule consists of.....bonds.

(A) One sigma and two pi
(B) One pi and two sigma
(C) Three pi
(D) Three sigma

Q6. If a gas having initial volume of $2dm^3$ and pressure of 2atm, what will be its new volume when the pressure is increased to 4 atm?

(A) 1 dm^3
(B) 2 dm^3
(C) 1/2 dm^3
(D) 4 dm^3

Q7. The process of diffusion and effusion are to be understand best by.....

(A) Boyle's law
(B) Charle's law
(C) Dalton's law
(D) Graham's law

Q8. Vacuum distillation is carried out under very reduced.....

(A) Pressure
(B) Temperature
(C) Volume
(D) Voltage

Q9. If NaCl is crystallized from a solution having 10 % urea as impurity in it, it forms.....crystals.

(A) Cubic
(B) Oval
(C) Round
(D) Needle like

Q10. Which one of the following is the strongest intermolecular force?

(A) Covalent bond
(B) Hydrogen bonding
(C) Dipole-dipole forces
(D) London dispersion

Q11. In which of the following reactions, the increase in pressure favours the products.

(A) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
(B) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
(C) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(D) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Q12. The ionization of NH_4OH is suppressed by adding.....

(A) NaCl
(B) HCl
(C) NH_4OH
(D) NH_4Cl

Q13. A buffer solution made by mixing a weak base and its salt with a strong acid has pH value.....

(A) More than 7
(B) Less than 7
(C) Equal to 7
(D) Equal to zero

Q14. If the rate of a chemical reaction is expressed as Rate = K [A] $[B]^n$, then overall order of reaction is...

(A) n
(B) 1
(C) n + 1
(D) n − 1

Q15. The process in which the water molecules surround and interact with the ions of solute is called....

(A) Neutralization
(B) Hydration
(C) Hydrolysis
(D) Hydrogenation

Q16. Which one of the given is not a colligative property?

(A) Lowering of vapour pressure
(B) Elevation of boiling point
(C) Depression in freezing point
(D) Viscosity

Q17. Enthalpy of combustion of a substance is.....

(A) Always negative
(B) Always positive
(C) Sometime positive or sometime negative
(D) Always zero

Q18. A cell which produces electric current by redox reaction is called a/an.....

(A) Voltaic cell
(B) Electrolytic cell
(C) Half cell
(D) Standard cell

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Answer any TEN questions. Each question carries FOUR marks.

Q1. What is the volume (in dm^3) of 2.5 moles of Oxygen gas (O_2) at STP?

Q2. Discuss the importance of dipole moment in determining the geometry of molecules?

Q3. What is Pauli Exclusion Principle? Explain it with an example.

Q4. Dalton's law of partial pressure is applicable for non-reacting gases only. Comment over it?

Q5. The strength of London dispersion forces depend upon the shapes of molecules. Explain it with reason and also give examples.

Q6. Explain the following with reasons: (a) Covalent crystals are hard while molecular crystals are soft. (b) Ionic solids are brittle

Q7. Equilibrium constant, Kp at 300 k is 0.75. Find Kc at the same temperature for the given reaction. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Q8. Write chemical equations for the following and also point out the Lewis acids and Lewis bases from the following: (a) Boron trifluoride and Ammonia (b) Ammonia and H^+

Q9. Enlist any four applications of Osmosis?

Q10. How does a catalyst increase the rate of a reaction? Compare a catalyzed and un-catalysed reaction on a potential energy diagram?

Q11. Write a note on Daniel cell?

Q12. Define internal energy of a chemical system? What are the possible consequences upon the system when internal energy of a system has increased?

Q13. Differentiate between the following terms: (a) Transition state and activated complex (b) Boiling point and evaporation

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

Q14.

(a) [5] Draw molecular orbital energy diagram for Oxygen molecule (O_2) and also calculate its bond order? What informations are obtained from its molecular orbital energy diagram and bond order?

(b) [4] Classify the colloids on the basis of interaction between dispersed medium and dispersion medium.

Q15.

(a) [5] Define quantum numbers. Give the importance of Azimuthal & Magnetic quantum Number.

(b) [4] Why do the gases show deviation from ideal behavior at high pressure & low temperature?

Q16.

(a) [2+3] Differentiate between solubility product constant and ionic product of an electrolyte. How the comparison of values of ionic product (IP) and solubility product constant (Ksp) help to know whether the solution is saturated, unsaturated or supersaturated?

(b) [4] What is anisotropy? Explain it with example?

Q17.

(a) [5] Derive a relationship between acid ionization constant (Ka) and ionization constant of its conjugate base (Kb)?

(b) [2+2] Balance the following equations by ion electron method. (i) $I^- + NO_2 \rightarrow I_2 + NO$ (Acidic Medium) (ii) $Zn + Cr_2O_7^{2-} + H^+ \rightarrow Zn^{2+} + Cr^{3+} + H_2O$ (Acidic Medium)